CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 10: **Tissues in Action** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The epithelium of the intestine is: 1

(a) cuboidal **(b)** columnar
(c) stratified squamous **(d)** squamous

2. The muscles of the heart are: 1

(a) voluntary muscles **(b)** skeletal muscles
(c) smooth muscles **(d)** cardiac muscles

3. The increase in the girth of the stem is due to the: 1

(a) lateral meristem (cambium) **(b)** intercalary meristem
(c) apical meristem **(d)** epidermis

4. Which of the following is a living component of xylem? 1

(a) xylem parenchyma **(b)** vessels **(c)** tracheids **(d)** fibres

5. The epithelium of the kidney tubules is: 1

(a) squamous **(b)** columnar **(c)** ciliated **(d)** cuboidal

6. The functional unit of nervous tissue is the: 1

(a) alveolus **(b)** neuron **(c)** osteocyte **(d)** nephron

7. Apical meristem is found at the: 1

(a) base of leaves **(b)** bark
(c) pith **(d)** tips of roots and shoots

8. The components of phloem are: 1

(a) cork cells
(b) guard cells
(c) sieve tubes, companion cells, phloem fibres and phloem parenchyma
(d) tracheids and vessels

9. The skin has: 1

(a) glandular epithelium **(b)** stratified squamous epithelium
(c) cuboidal epithelium **(d)** columnar epithelium

10. The three parts of a neuron are: 1

(a) xylem, phloem, cambium **(b)** dendrites, cell body and axon
(c) nucleus, wall, vacuole **(d)** head, neck, tail

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What are the functions of stomata? 2

12. What is the difference between a tendon and a ligament? 2

13. Why do meristematic cells lack large vacuoles? 2

14. Name the three types of muscle tissue. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe the structure of xylem and its functions. 3

16. State the functions of blood, bone and adipose tissue. 3

17. What is tissue culture? State two uses. 3

OR

Describe the structure of phloem and its functions. 3

18. Why does grass regrow after being eaten by grazing animals? 3

OR

Why are muscle cells long and nerve cells branched? Relate structure and function. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the complex tissues xylem and phloem with their elements and functions. 5

OR

Describe the joints, bones, cartilage and muscle together as a system for movement with examples. 5

20. Describe the protective tissues in plants: epidermis, cuticle, stomata, root hairs and cork. 5

OR

Explain the importance of tissue culture and the experiment by F. C. Steward with its applications. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND